

Front wing in ground effect beside a rotating wheel

Steady RANS mesh and setup report

OpenFOAM v2606 (native) • snappyHexMesh + simpleFoam • case S1.42_h0.13_AOA8_W0.63

Scope of this report. This is a **mesh and setup** report. The geometry, the domain, the mesh and the solver configuration are complete and verified; the numbers in it — cell counts, refinement distribution, layer coverage, **checkMesh**, meshing time — are mesh-side quantities and can be relied on. **No flow solution exists.** The case was meshed and stopped there, so there are no forces, no C_p , no y^+ and no convergence history to report. The physics and numerics in section 6 describe what *is configured*, not what has been run. Next step: section 7.

1 Case at a glance

Configuration		Mesh	
wing half-span	$S/c = 1.42$	cells	18 322 109
ride height	$h/c = 0.13$	background cells	123 624
angle of attack	8°	layer coverage	96.0 %
wheel width	$W/c = 0.63$	checkMesh clean	no
Flow it is built for		Levels	
freestream	10.0 m/s	base cell	18.75 mm = $0.25c$
chord	0.075 m	L4	1.17 mm
Re_c	46 793	L5	0.586 mm
ρ	1.1629 kg/m ³	L6	0.293 mm
ν	1.60278e-05 m ² /s	L7	0.146 mm

Meshing took 20 min on 20 MPI ranks. No solve has been run.

What the forces will be reported as, once the case is solved. Force *coefficient-areas*, not coefficients: $A_{\text{ref}} = 1 \text{ m}^2$ is set on purpose in **system/forceCoeffs**, so **forceCoeffs** writes $C_L \cdot A$ and $C_D \cdot A$ in m² — the quantity the Diasinos papers tabulate, and the one that sidesteps which reference area to use as S/c varies across the sweep. Negative lift is downforce. Three groups are configured: wing, wheel and both together.

2 Software and reproducibility

- **OpenFOAM v2606** (ESI/OpenCFD), built from source. Native only, no third-party CFD tooling.
- Mesh **snappyHexMesh**, solve **simpleFoam** (segregated, SIMPLEC), turbulence **kOmegaSST**.
- Geometry generated parametrically from the Diasinos specification; 420 configurations available.
- Everything else is **python3** standard library. No **numpy/pandas** needed to run a case.

To reproduce the mesh in this report.

```
cp env/project.env.example env/project.env # set OPENFOAM_BASHRC and NP
cd Diasinos_Geometry_Generator && .venv/bin/python -m geometry_generator.generate
cd ../RANS_Simulations
python3 prepare_case.py --span 1.42 --height 0.13 --aoa 8 --width 0.63
cd S1.42_h0.13_AOA8_W0.63 && ./Allmesh
```

3 Geometry

- Coordinate system: x streamwise (wheel centre at $x = 0$), y spanwise outboard, z up with the ground at $z = 0$. Half model, symmetry at $y = 0$.
- Chord $c = 75$ mm. Inverted NACA 4412, blunt trailing edge 0.5 mm. Wheel diameter $1.17c$, shoulder radius $0.067c$.
- **The generator writes millimetres**; everything in the simulation is metres. **Allmesh** scales by 0.001 on the way in.
- The *plinth* is a collar that deliberately pierces the ground plane (down to $z = -6$ mm) so the contact patch is a clean cut instead of a tangential singularity.

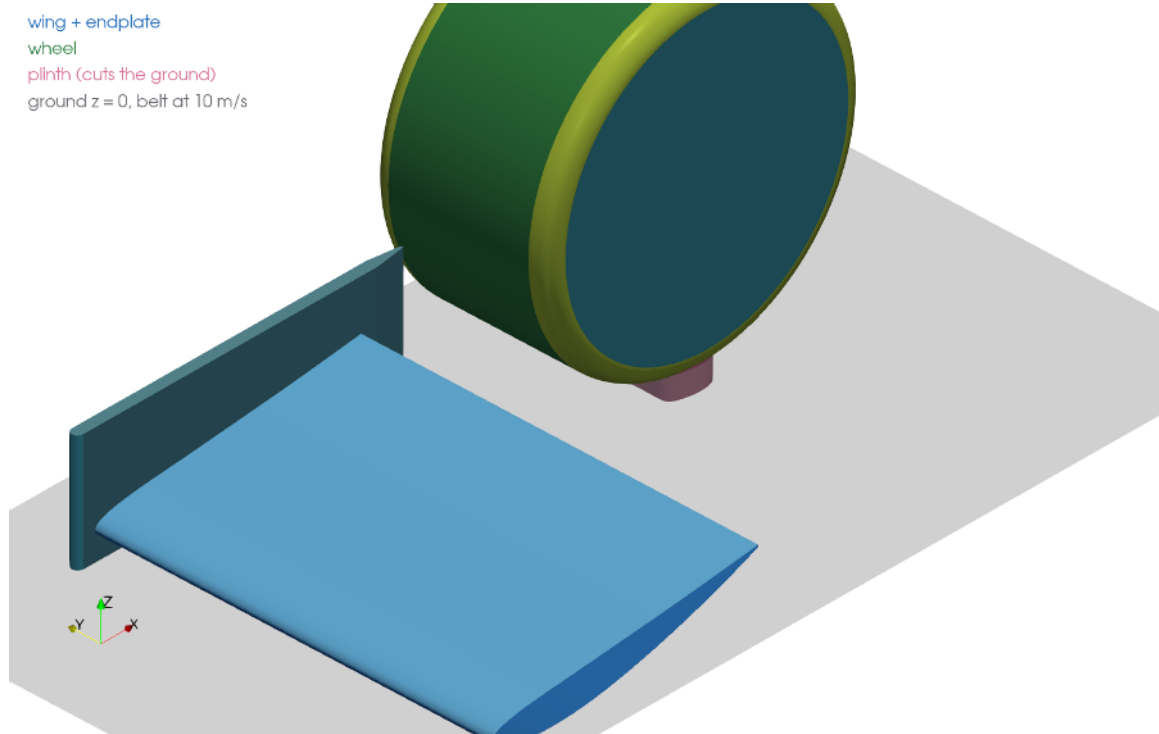


Figure 1: Geometry and the ground plane. The wing sits in ground effect ahead of and inboard of the wheel; the plinth passes through the ground.

Known geometry limitations.

- The wing is **prismatic**, extruded between two spanwise stations. Geometrically exact for a constant section; it does mean the triangles have a very high aspect ratio, which **surfaceCheck** reports as poor quality. Not an error.
- **Two duplicate triangles** on the endplate outer edge — a defect in the generator. Two facets out of 1142; **snappyHexMesh** tolerates them. Reported by **Allmesh** and deliberately non-fatal.
- Both STLs consist of two disjoint parts (wing + endplate, wheel + plinth), so **surfaceCheck** says “not closed”. Expected: no **mode inside** refinement region is attached to either body.
- Wheel tessellation was raised from 180 to 900 circumferential segments, so the facet is 0.31 mm against the 0.293 mm cell. At 180 segments the facets were 1.53 mm — five times coarser than the cell, i.e. the mesh resolved the wheel as a 180-gon with a visibly wavy surface. It was done for geometric fidelity; the tread now carries 3.95 of 4 layers at 84.3 % of the requested thickness (table above).

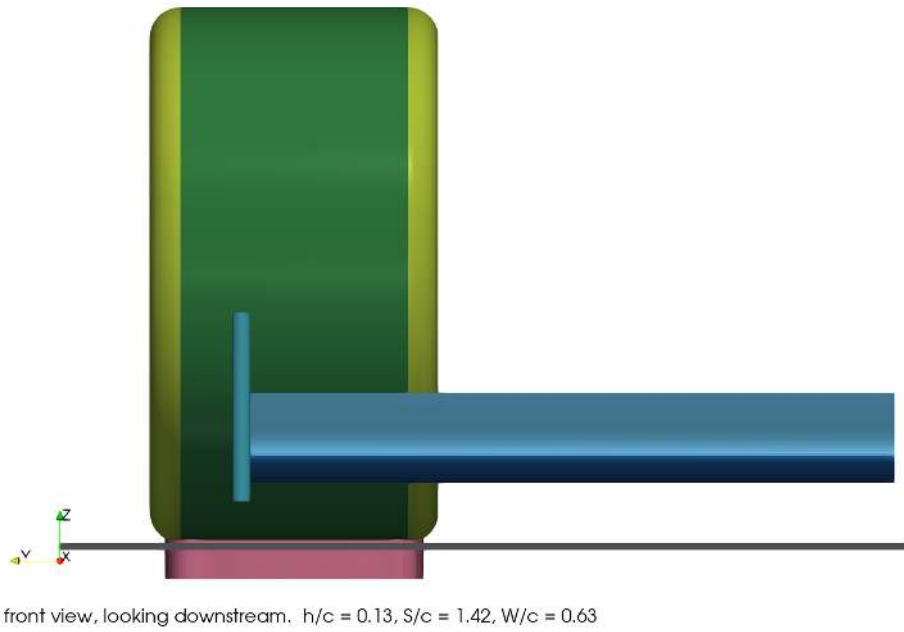


Figure 2: Front view, looking downstream. The endplate (light blue) falls inside the wheel silhouette: at $S/c = 1.42$ and $W/c = 0.63$ the spanwise gap is $-0.45c$, i.e. wing and wheel overlap.

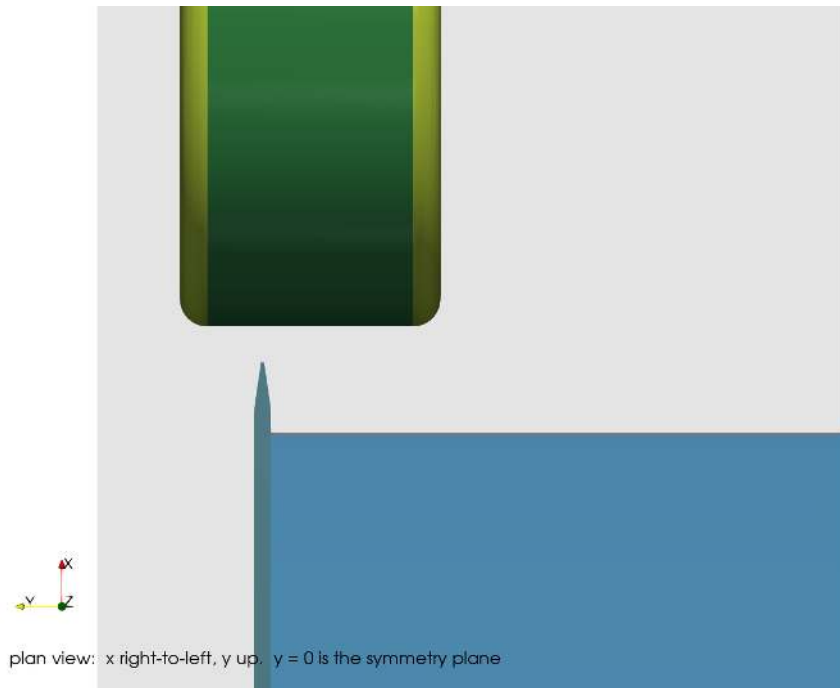


Figure 3: Plan view. $y = 0$ is the symmetry plane.

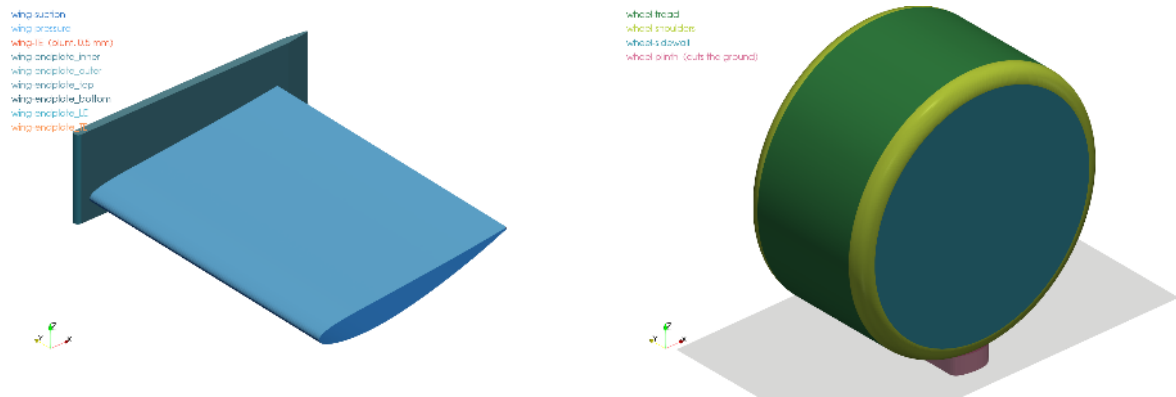


Figure 4: Named STL regions. `snappyHexMeshDict` refers to these names directly to set refinement level and patch name, so renaming a region in the generator breaks the mesh setup.

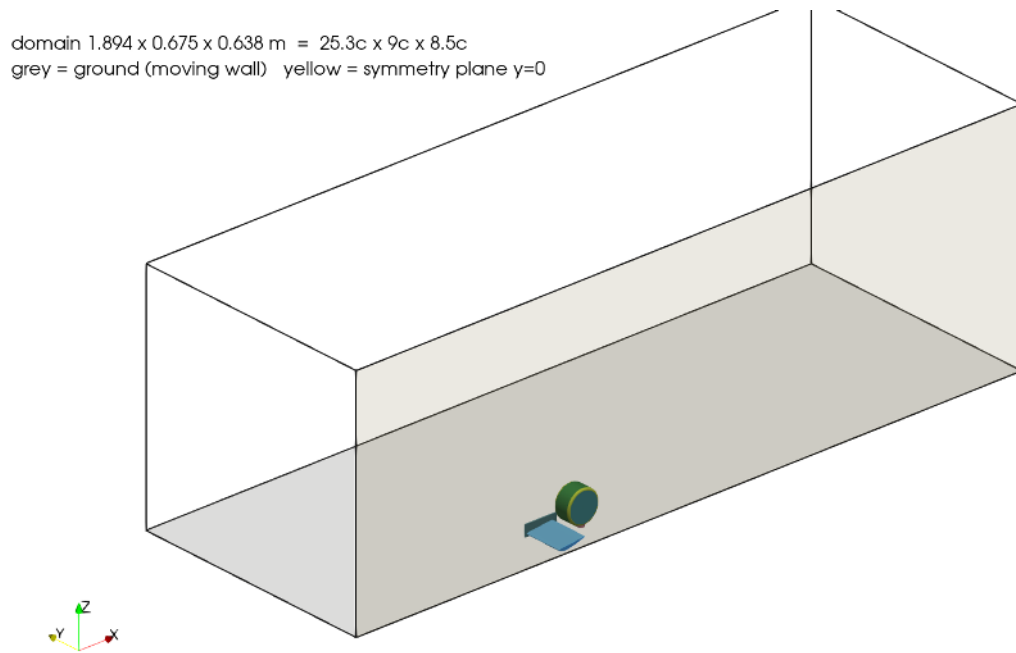


Figure 5: Computational domain, $1.894 \text{ m} \times 0.675 \text{ m} \times 0.638 \text{ m} = 25.3c \times 9c \times 8.5c$. Grey: moving ground. Yellow: symmetry plane.

4 Domain and boundary conditions

patch / group	type	U	p	k, ω, ν_t
inlet	patch	fixedValue (1000)	zeroGradient	turbulent inlet
outlet	patch	pressureInletOutletVelocity	fixedValue 0	inletOutlet
symmetry	symmetry	symmetry	symmetry	symmetry
side, sky	patch	slip	slip	slip / calculated
ground	wall	fixedValue (1000)	zeroGradient	wall functions
wingGroup	wall	noSlip	zeroGradient	wall functions
wheelRotating	wall	rotatingWallVelocity	zeroGradient	wall functions
wheelGround	wall	fixedValue (1000)	zeroGradient	wall functions

- **Rolling road:** the belt translates downstream at the freestream speed, so the ground is at rest relative to the far-field air.
- **Rotating wheel:** $\omega = -227.92$ rad/s about (0 1 0) through (0, 0, 0.043875 m). `rotatingWallVelocity` imposes $\omega \times r$ without moving the mesh — correct for steady RANS. The sign follows from requiring the contact point to move with the belt: $\omega_y = -U_\infty/R$. Reversing it spins the wheel backwards and still converges happily.
- The **wheel-plinth** patch is in `wheelGround`, not `wheelRotating`: it is the collar through the ground and moves with the belt.
- **side** and **sky** are `patch`, not `wall`, on purpose — a wall there would enter the `meshWave` wall-distance field that `kOmegaSST` blends on, and would appear in the force groups.

5 Mesh

- Background: $101 \times 36 \times 34$ **cubic** cells of 18.75 mm ($= 0.25c$), 123 624 cells. Cubic matters: each refinement level halves the edge, so the level-to-size table only holds from a cube.
- Level sizes: L4 1.17 mm, L5 0.586 mm, L6 0.293 mm, L7 0.146 mm.
- Per-region surface levels, distance refinement around both bodies (L5 within 15 mm, L4 within 30 mm), and nine `searchableBox` zones.
- Final mesh 18 322 109 cells.

refinement level	cells	share [%]
L0	110 876	0.6
L1	19 040	0.1
L2	519 518	2.8
L3	592 509	3.2
L4	4 302 360	23.5
L5	11 579 533	63.2
L6	1 194 917	6.5
L7	3 356	0.0

5.1 Streamwise stations through the front wing

The y - and z -slices above cut the wheel; neither shows what the mesh does *around the wing and its endplate*, which is where the vortex that drives this whole study is created. Two x -normal stations do. Both look downstream, so y grows to the left and the symmetry plane is at the right edge.

5.2 Prism layers

Four layers from 0.14 mm with expansion 1.3 (total 0.87 mm) on wing and wheel, three from 0.25 mm on the ground. This targets $y^+ \approx 2.5$, which is why the wall functions are the all- y^+ variants.

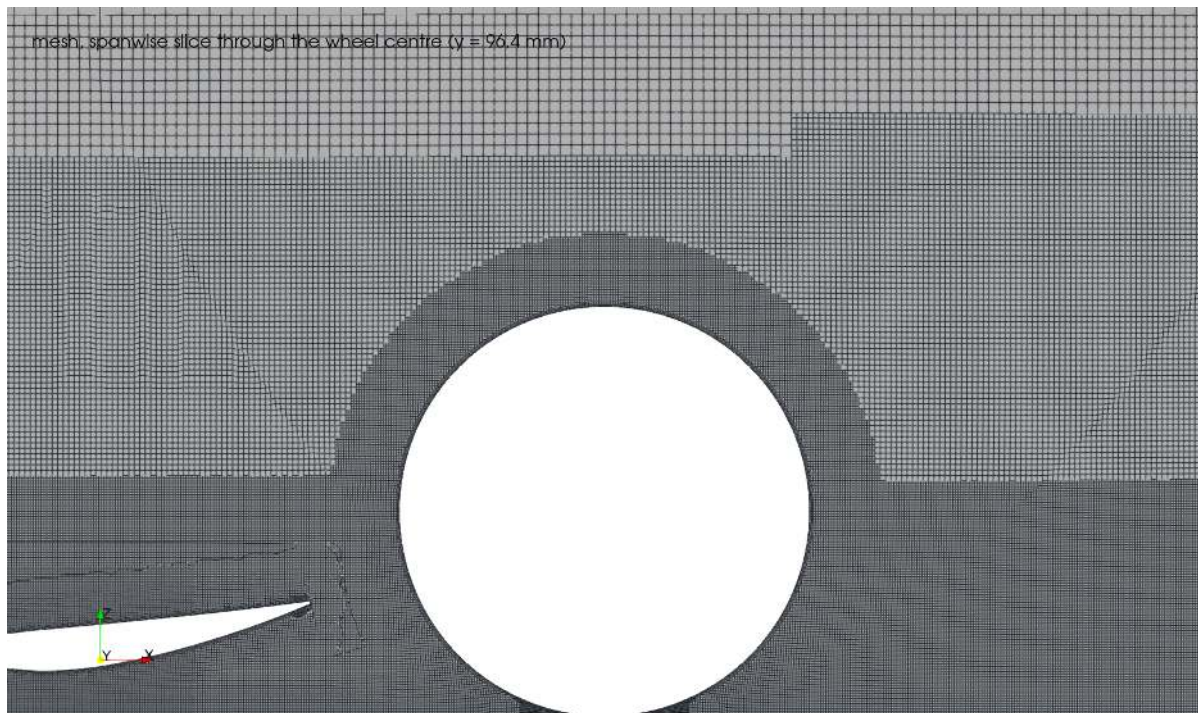


Figure 6: Mesh on a spanwise slice through the wheel centre. The nested refinement zones and the wake box are visible.

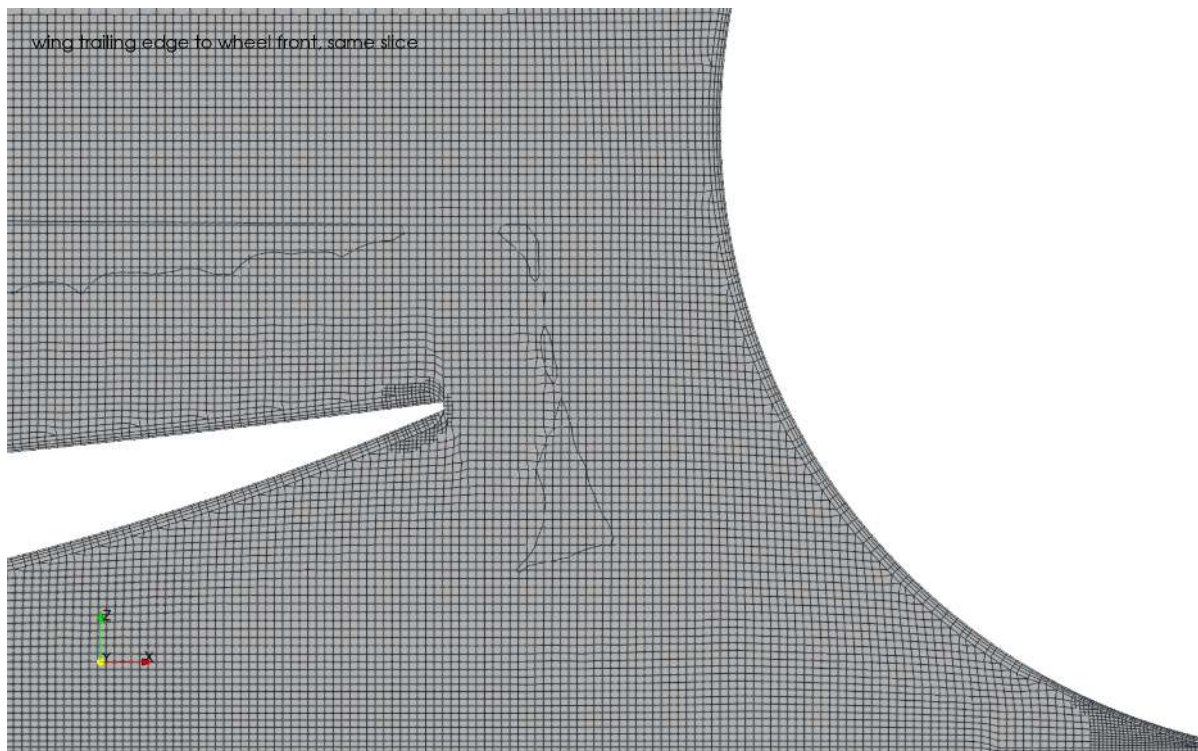


Figure 7: Same slice, wing trailing edge to wheel front — the region the whole study is about.

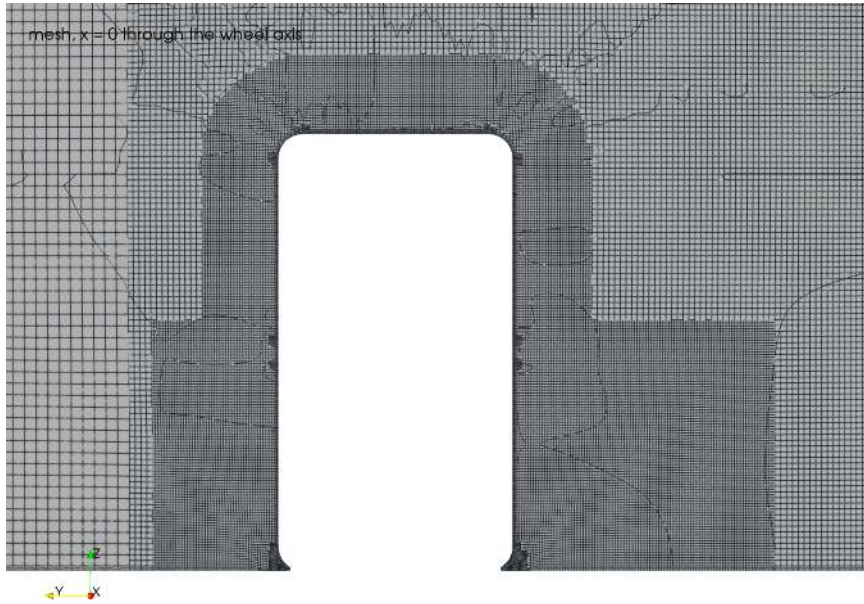


Figure 8: Slice through the wheel axis ($x = 0$). Looking downstream, so y increases to the left.

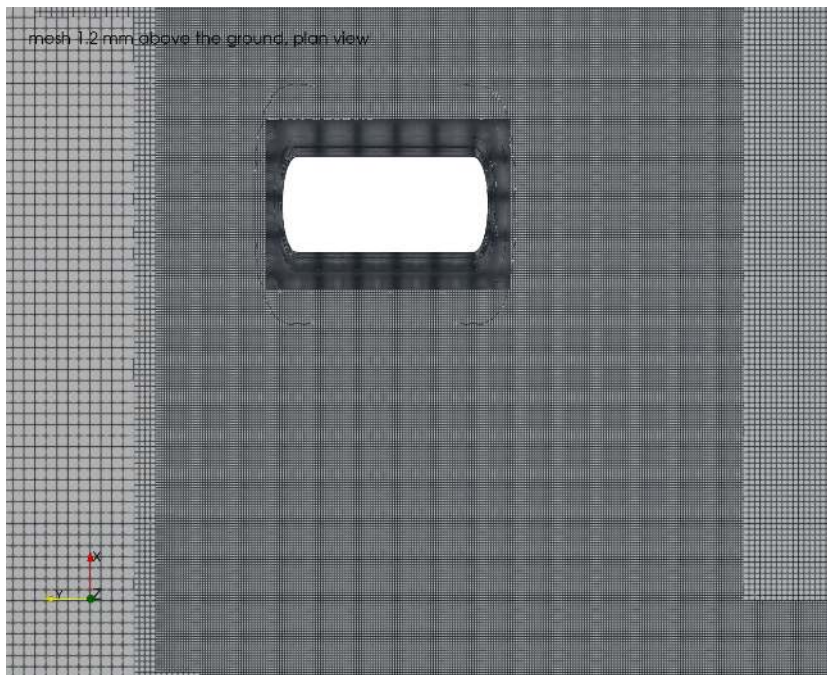


Figure 9: Mesh 1.2 mm above the belt. The contact-patch refinement box is the fine square under the wheel.



Figure 10: $x = -95$ mm, through the wing and the full endplate — roughly mid-chord. The wing section is the horizontal void, the endplate the vertical one. Visible here: the L5–L6 shell that the wing’s own surface levels produce, the L4/L5 distance refinement stepping outwards from both, and the fine patch wrapped around the endplate bottom edge at the ground. That last one comes from the `endplateBottomEdge` geometry entry (L6 within 6 mm), not from a box, which is why it follows the edge across the parameter sweep.

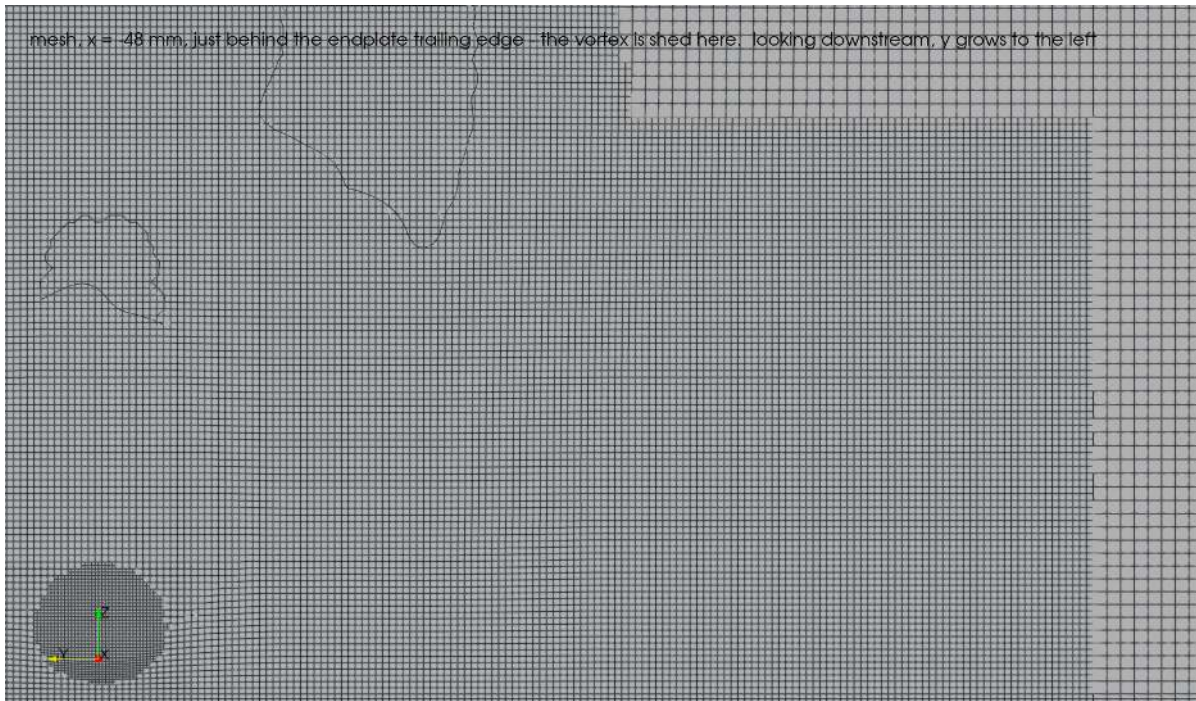


Figure 11: $x = -48$ mm, just behind the endplate trailing edge at -50.4 mm — the station where the bottom-edge vortex is shed. No geometry is cut any more, so what is left is purely refinement: the L6 core at the ground where the vortex forms, and the L5-vortex-inboard/outboard boxes carrying it downstream. This is the figure to check when the parameter range is extended, because those two boxes are fixed and do not follow the geometry.

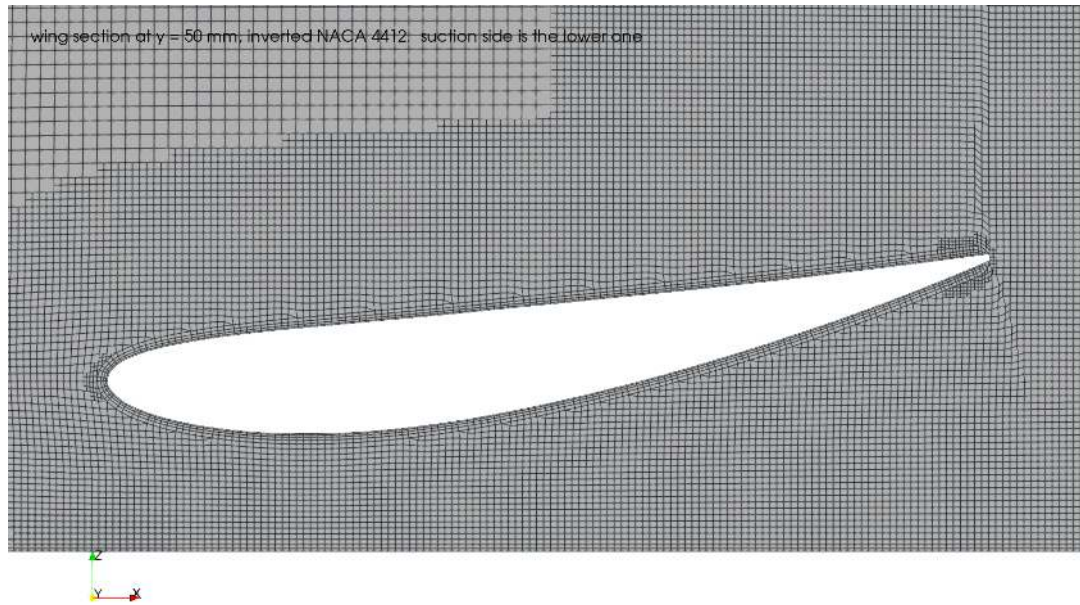


Figure 12: The wing section itself, on a spanwise slice at $y = 50$ mm: inverted NACA 4412, suction side underneath. For scale, the chord is 75 mm and the cells on the surface are 0.586 mm (L5) escalating to 0.293 mm (L6).

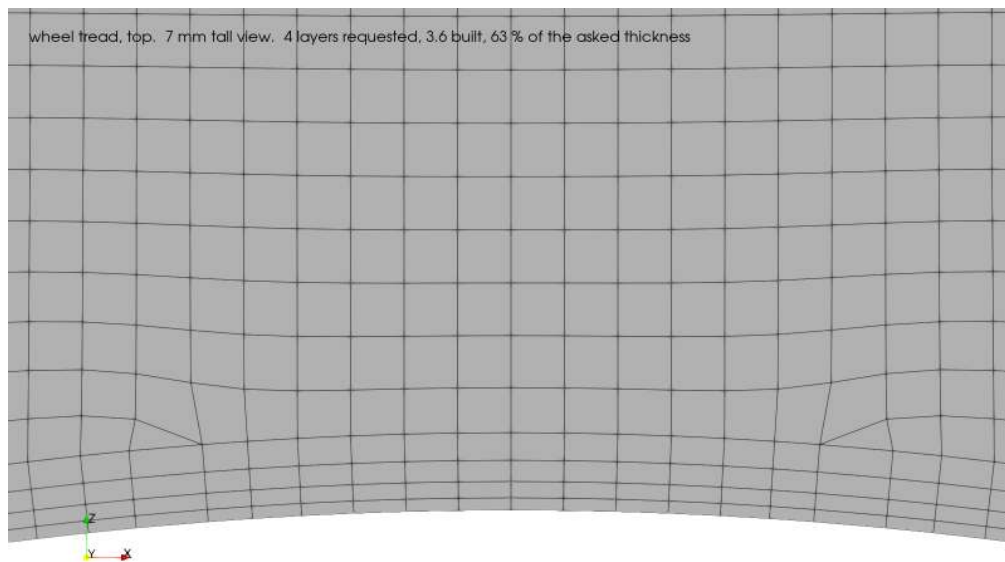


Figure 13: Wheel tread at the top. Prism layers on the 0.31 mm-tessellated tread.

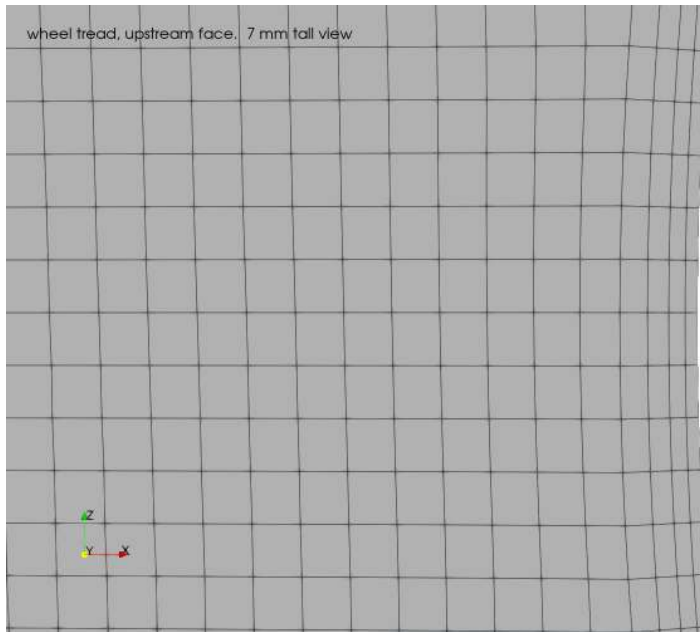


Figure 14: Wheel tread, upstream face.

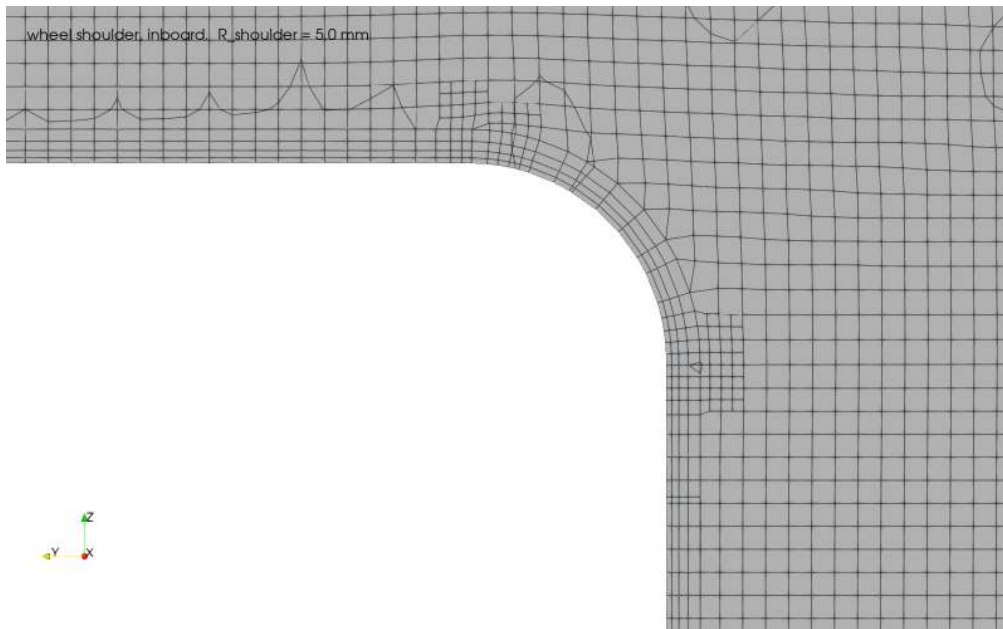


Figure 15: Wheel shoulder, inboard. The 5 mm shoulder radius is the tightest convex curvature on the wheel.

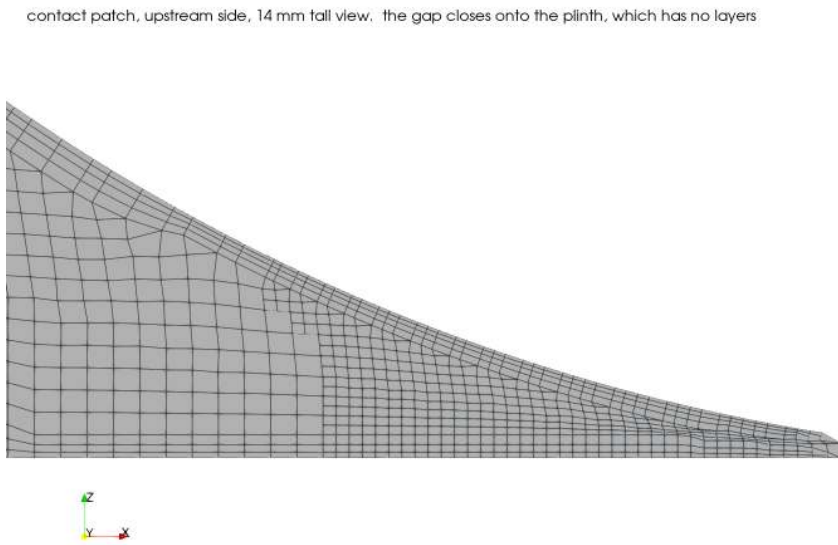


Figure 16: Contact patch. The plinth cuts the ground plane at $z = 0$; this is where the layers fight each other.

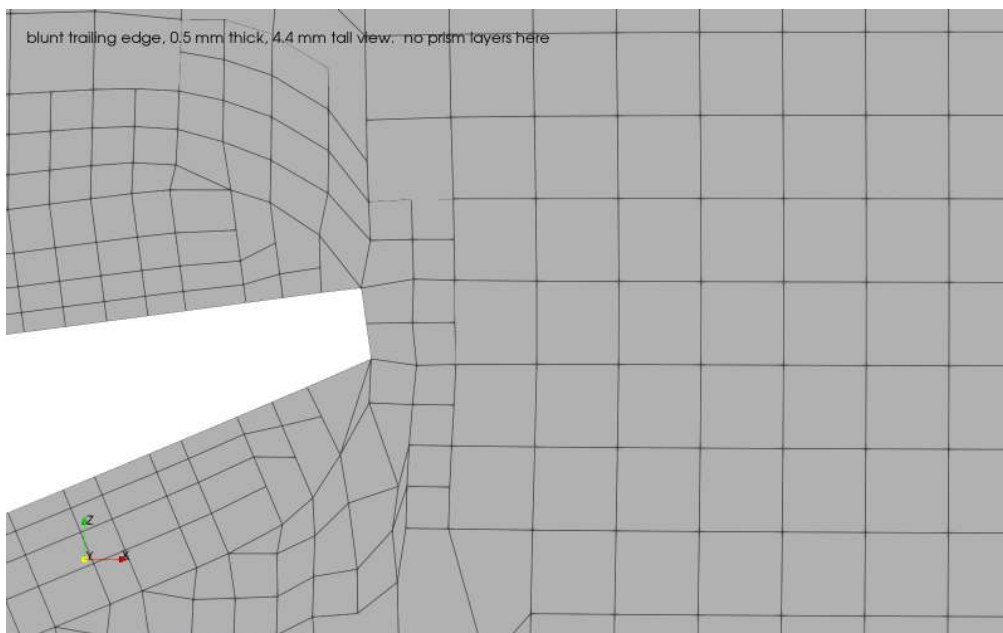


Figure 17: Blunt trailing edge, 0.5 mm thick. No prism layers — see below.

patch	faces	layers wanted	layers built	thickness [%]
ground	180851	3	2.92	92.3
wing-suction	28951	4	3.88	95.1
wing-pressure	27637	4	3.87	96.2
wing-TE	746	2	0	0
wing-endplate_inner	13234	4	3.45	79.1
wing-endplate_outer	14498	4	3.71	94.3
wing-endplate_top	3256	4	2.85	76
wing-endplate_bottom	3256	4	2.85	76.4
wing-endplate_LE	1356	4	3.87	97.3
wing-endplate_TE	212	2	0	0
wheel-tread	38861	4	3.95	84.3
wheel-shoulders	26056	4	3.88	89.9
wheel-sidewall	33128	4	4	100
wheel-plinth	902	2	0	0

Reading the layer table.

- Overall coverage 96.0 %.
- The patches in **bold** got **no layers at all**. These are **wing-TE**, **wing-endplate_TE** and **wheel-plinth** — sharp convex edges and the plinth/ground intersection. Asking for fewer and thinner layers does not change it: the limit is topology, not thickness.
- **This is accepted.** `snappyHexMesh` is known for incomplete layer coverage at sharp edges and where two layered surfaces meet. **wing-TE** and **wheel-plinth** carry surface level (6 7), so the cell at the wall there is 0.293 mm at its coarsest and the near-wall spacing is comparable to the layered regions anyway. **wing-endplate_TE** is the weak one: level (5 5), so 0.586 mm cells with no layers — but it is 212 faces on the endplate trailing edge, away from the pressure-carrying surfaces.
- The number to judge the near-wall treatment by is y^+ , not the layer count — which means it cannot be judged yet. It is the first thing to check once the case is solved (section 7).

5.3 Mesh quality

Quality limits are deliberately relaxed (`maxNonOrtho` 75 against a default of 65, `maxInternalSkewness` 6 against 4, `minTetQuality` -10^{30} , which disables the constraint). Enforcing a positive tet quality with absolute layer thicknesses on curved surfaces costs most of the layer coverage.

`checkMesh` clean: **no** (4 findings).

- Error in face tets: 248 faces with low quality or negative volume decomposition tets.
- Cells with small determinant (< 0.001) found, number of cells: 531
- Concave cells (using face planes) found, number of cells: 154366
- Faces with small interpolation weight (< 0.05) found, number of faces: 80

These are the expected consequence of the relaxed limits and are accepted in exchange for layer coverage. If the solve later diverges immediately, `meshQualityDict` is the first thing to question.

6 Physics and numerics as configured

Nothing in this section has been run. It is the solver setup the mesh was built for, recorded here so the mesh choices — the layer stack in particular — can be judged against it.

turbulence model	kOmegaSST
ν_t wall function	nutUSpaldingWallFunction (continuous all y^+)
k wall function	kLowReWallFunction
ω wall function	omegaWallFunction
momentum convection	bounded Gauss linearUpwindV grad(U)
turbulence convection	bounded Gauss upwind (first order, deliberate)
gradients	cellLimited Gauss linear 1
laplacian	Gauss linear corrected
wall distance	meshWave
pressure-velocity	SIMPLEC (consistent yes)
p solver	GAMG, tolerance 10^{-7} , relTol 0.01
U, k, ω	smoothSolver GaussSeidel
relaxation	U 0.9, k and ω 0.7
residualControl	p, U : 10^{-5} ; k, ω : 10^{-4}
initialisation	potentialFoam, divergence-free start

Turbulence convection is first order on purpose. Second-order k/ω transport is the usual reason residuals stall on a mesh of this kind, and the effect on the integrated forces is small. Momentum convection is second order.

Two model limitations to carry forward. Native OpenFOAM's kOmegaSST has neither of these, so both are absent by necessity, not by choice:

- **No curvature correction.** The endplate and wheel vortex cores will diffuse somewhat faster than a curvature-corrected model would give.
- **No separation fix.** Separation onset on the suction side may differ.

The near-wall treatment is covered by the all- y^+ wall functions above, and the startup by potentialFoam initialisation.

7 Next step: the solve

The mesh is finished and the case is configured; the solve has not been started. From the meshed case:

```
cd RANS_Simulations/S1.42_h0.13_AOA8_W0.63
./Allsolve                # potentialFoam, then simpleFoam
./Allpost --summary       # writes results.json / results.txt

cd ../../report
./build.sh                # regenerates the figures, rebuilds the PDF
```

- ./Allsolve is resumable: startFrom is latestTime in controlDict, so a second call continues rather than restarting.
- The run stops by itself once residualControl is met (p, U : 10^{-5} ; k, ω : 10^{-4}), which should be well before the configured 2000 iterations.
- To stop it cleanly and keep the state resumable, set stopAt writeNow in system/controlDict — the solver writes the current state and exits. runTimeModifiable is on, so it picks the change up within one iteration. Killing the process instead loses everything since the last write.
- build.sh -no-figures rebuilds only the text, which is quick.

What to check first, once there is a solution. In this order: y^+ against the design value of ≈ 2.5 the layer stack was sized for; then the spread of $C_L \cdot A$ over the averaging window, which matters more than the residual — a converged residual with the forces still swinging by several percent means an unsteady flow being forced into a steady solve.

Nothing in this report is typed by hand: every number comes from the case's logs via `report/scripts/make_tex_` so a rebuild picks the mesh up automatically.

8 Assessment

What supports trusting this mesh.

- **Background mesh exactly as designed** (123 624 cells), cubic, dividing the domain without remainder — which is what makes the level-to-cell-size table hold.
- **Refinement distribution is as intended**, level for level, with the share per level in the table in section 5 and no level running into the `maxGlobalCells` cap.
- **Patch groups resolve correctly** — `patchSummary` confirms `noSlip` on the wing, `rotatingWallVelocity` on the wheel and the belt velocity on the plinth, so the boundary conditions address the patches they are meant to.
- **Wheel rotation sign verified** analytically against the belt speed.
- **Endplate-edge refinement follows the geometry**, via the extracted `endplateBottomEdge` surface rather than a fixed box, so it holds across the parameter sweep.

What to be careful about.

- **Relaxed mesh quality limits** leave genuinely poor cells in the mesh (`minTetQuality` disabled). Accepted in exchange for layer coverage, but it is the first thing to question if the solve diverges.
- **Three patches without prism layers**, see section 5.
- **Two duplicate triangles** in the wing STL.
- **No curvature correction and no separation fix** in the configured turbulence model. Expect faster vortex diffusion and possibly a different separation onset than a model with both would give.
- **Steady RANS on a flow that may not be steady**. At high angle of attack or low ride height, check the force spread before quoting any number the solve produces.
- **L5-vortex-inboard** and **L5-vortex-outboard** are **fixed boxes positioned for this geometry** and do not follow the parameters. Re-check them before extending the sweep.
- **No solution, so no y^+ yet**. The layer stack is sized for $y^+ \approx 2.5$ (section 5), but that is a design target, not a measurement. It is the one number that can retrospectively invalidate the near-wall part of this mesh.